

Project Report: Performance + Security Test Program - QA & Test Engineer

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Family:	QA & Test Engineer
Level:	Advanced
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

k6 load test plan, OWASP ZAP scan, and combined quality gate report for a web application.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Advanced]

Chapter focus: Project Overview** *(Level: Advanced)

Performance + Security Test Program** is a advanced-level end-to-end project for QA & Test Engineer. **Duration:** 4-5 weeks. k6 load test plan, OWASP ZAP scan, and combined quality gate report for a web application. **Tools:** k6, OWASP ZAP, GitHub Actions.

Code Reference:

```
Objective: k6 load test to 500 virtual users
Objective: OWASP ZAP automated scan
Objective: Performance baseline document
Objective: Combined quality gate checklist
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

QA & Test Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a QA & Test Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Load [Advanced]

Chapter focus: Phase: Load** *(Level: Advanced)

k6 scripts. Complete these tasks: Ramp up/down; Thresholds. Deliverable for this phase: k6 HTML report.

Code Reference:

- [] Ramp up/down
- [] Thresholds

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Load is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

QA & Test Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the k6 HTML report to the teacher.

Chapter 3: Phase: Security [Advanced]

Chapter focus: Phase: Security** *(Level: Advanced)

ZAP baseline scan. Complete these tasks: Top 10 checks; Remediation list. Deliverable for this phase: ZAP report.

Code Reference:

- [] Top 10 checks
- [] Remediation list

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Security is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

QA & Test Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the ZAP report to the teacher.

Chapter 4: Phase: Gate [Advanced]**Chapter focus: Phase: Gate** *(Level: Advanced)**

Quality criteria. Complete these tasks: p95 < 500ms; Zero critical vulns. Deliverable for this phase: Gate checklist.

Code Reference:

- ☐ p95 < 500ms
- ☐ Zero critical vulns

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Gate is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

QA & Test Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Gate checklist to the teacher.

Chapter 5: Phase: Present [Advanced]**Chapter focus: Phase: Present** *(Level: Advanced)**

QA sign-off deck. Complete these tasks: Findings; Go/no-go. Deliverable for this phase: Sign-off presentation.

Code Reference:

- ☐ Findings
- ☐ Go/no-go

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Present is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

QA & Test Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Sign-off presentation to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Advanced]

Chapter focus: Final Deliverables and Review *(Level: Advanced)**

Submit all final deliverables: k6 report; ZAP report; Quality gate doc; Sign-off deck. Skills demonstrated: Performance, Security, Quality gates. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

Deliverable: k6 report
Deliverable: ZAP report
Deliverable: Quality gate doc
Deliverable: Sign-off deck

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide**What it actually is**

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits k6 report and passes the rubric.